

DataFrame_CSV

May 25, 2026

1 Working with DataFrame and CSV

CSV stands for **Comma separated Values**. These files are easy to manipulate, and can be used across multiple applications. Example: DataFrame, SQL, etc.

The default delimiter in CSV is comma `,`. But, we can use different delimiters in our CSV files. We have to mention the delimiter, except for comma.

```
[18]: # Import a CSV file into DataFrame
import pandas as pd

df = pd.read_csv("data.csv")
print(df)
```

	0	1	2
0	RollNo	Name	Age
1	101	Ramu	16
2	102	Sonu	19
3	103	Rita	16
4	104	Amit	20

By default, the first row of the CSV file is always treated as the **header row**.

```
[3]: # If we consider the second row as our header row
df = pd.read_csv('data.csv', header=1)
print(df)
```

	101	Ramu	16
0	102	Sonu	19
1	103	Rita	16
2	104	Amit	20

```
[4]: df = pd.read_csv('data.csv', header=2)
print(df)
```

	102	Sonu	19
0	103	Rita	16
1	104	Amit	20

Suppose the csv file does not have any header row, then:

```
[19]: df = pd.read_csv("data.csv", header=None)
      print(df)
```

	0	1	2
0	RollNo	Name	Age
1	101	Ramu	16
2	102	Sonu	19
3	103	Rita	16
4	104	Amit	20

This makes all the rows in our CSV file as a part of our DataFrame, and system assigns a new index to our columns.

```
[5]: df = pd.read_csv("data.csv")
     print(df)
```

	RollNo	Name	Age
0	101	Ramu	16
1	102	Sonu	19
2	103	Rita	16
3	104	Amit	20

```
[6]: df = pd.read_csv("data.csv", index_col='RollNo')
     print(df)
```

	Name	Age
RollNo		
101	Ramu	16
102	Sonu	19
103	Rita	16
104	Amit	20

Instead of putting a new system-defined index in our DataFrame, we are considering RollNo as our index values. This same code can be written as follows:

```
[7]: df = pd.read_csv("data.csv", index_col=0)
     print(df)
```

	Name	Age
RollNo		
101	Ramu	16
102	Sonu	19
103	Rita	16
104	Amit	20

If we make Age as our index value, then:

```
[8]: df = pd.read_csv("data.csv", index_col=2)
     print(df)
```

	RollNo	Name
--	--------	------

Age		
16	101	Ramu
19	102	Sonu
16	103	Rita
20	104	Amit

```
[9]: df = pd.read_csv("data.csv")
      print(df)
```

	RollNo	Name	Age
0	101	Ramu	16
1	102	Sonu	19
2	103	Rita	16
3	104	Amit	20

1.1 Filter the CSV file value:

Finding details of those students whose Age is more than 16.

```
[14]: age_filter = df[df['Age']>16]
      print(age_filter)
```

	RollNo	Name	Age
1	102	Sonu	19
3	104	Amit	20

```
[15]: print(type(age_filter))
```

```
<class 'pandas.DataFrame'>
```

1.2 Store the filtered DataFrame to a CSV file

```
[16]: age_filter.to_csv('filter_df.csv')
```

If we want to ignore the index values of the DataFrame, then:

```
[17]: age_filter.to_csv('filter_df.csv',index=False)
```

Setting index=False ignores the index which is present in the DataFrame.

```
[ ]:
```